LECTURE 6

Data Wrangling and EDA, Part II

Exploratory Data Analysis and its role in the data science lifecycle.

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Announcements

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Today’s Roadmap

Structure

- Multiple Files
- More File Formats

Scope and Temporality

Faithfulness (and Missing Values)

- Demo: Mauna Loa CO2

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Multiple Files

Structure

- **Multiple Files**
- More File Formats

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Key Data Properties to Consider in EDA

File Format

Variable Type

Multiple files
(Primary and Foreign Keys)

Structure -- the “shape” of a data file

Granularity -- how fine/coarse is each datum

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What is incidence?

Summary

What is already known about this topic?

The number of reported U.S. tuberculosis (TB) cases decreased sharply in 2020, possibly related to multiple factors associated with the COVID-19 pandemic.

What is added by this report?

Reported TB incidence (cases per 100,000 persons) increased 9.4%, from 2.2 during 2020 to 2.4 during 2021 but was lower than incidence during 2019 (2.7). Increases occurred among both U.S.-born and non–U.S.-born persons.

What are the implications for public health practice?

Factors contributing to changes in reported TB during 2020–2021 likely include an actual reduction in TB incidence as well as delayed or missed TB diagnoses. Timely evaluation and treatment of TB and latent tuberculosis infection remain critical to achieving U.S. TB elimination.

CDC Morbidity and Mortality Weekly Report (MMWR) 03/25/2022.

What is **incidence**? Why use it here?

How was "9.4% increase" computed?

Question: Can we **reproduce** these rates using government data?

CDC: source, 2021

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Defining incidence

From the CDC report: **TB incidence** is computed as the number of "cases per 100,000 persons using mid-year population estimates from the U.S. Census Bureau."

- Incidence is useful when comparing case rates across differently sized populations.

$$\textbf{TB incidence} = \frac{\text{\# TB cases in population}}{\text{\# groups in population}}$$

(group: 100,000 people)

$$= \frac{\text{\# TB cases}}{(\text{population}/100{,}000)}$$

$$= \frac{\text{\# TB cases}}{\text{population}} \times 100{,}000$$

We don't have U.S. Census population data in our DataFrame.
We need to acquire it to verify incidence!

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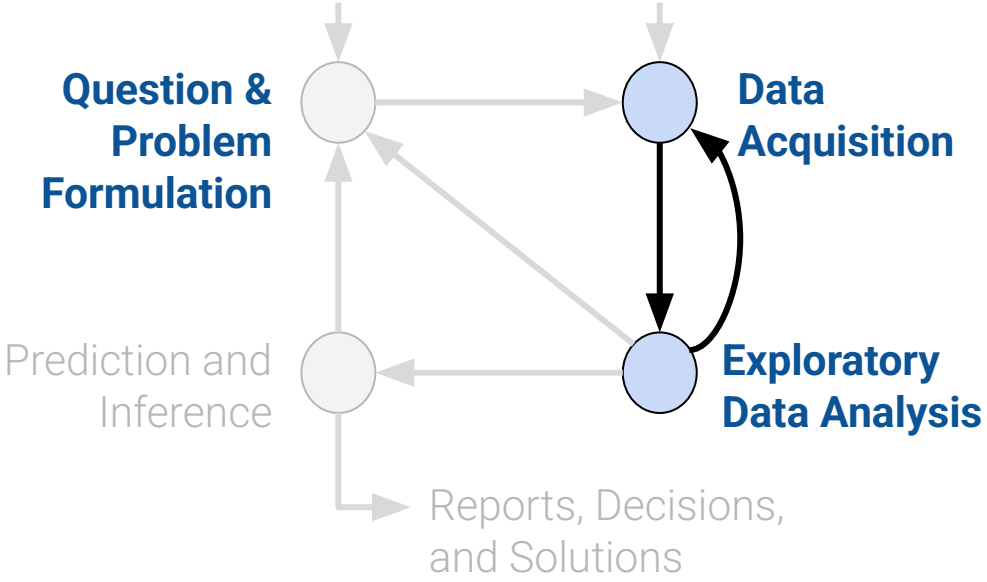
The Data Science Lifecycle is a Cycle

In practice, EDA informs whether you need more data to address your research question.



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Structure: Primary Keys and Foreign Keys

Sometimes your data comes in multiple files:

- Often data will reference other pieces of data.
- Alternatively, you will collect multiple pieces of related data.

Use `pd.merge` to **join** data on **keys**.

Customers.csv

CustID	Addr
171345	Harmon..
281139	Main ..

Orders.csv

OrderNum	CustID	Date
1	171345	8/21/2017
2	281139	8/30/2017

Products.csv

ProdID	Cost
42	3.14
999	2.72

Purchases.csv

OrderNum	ProdID	Quantity
1	42	3
1	999	2
2	42	1

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Primary key: the column or set of columns in a table that *uniquely* determine the values in the remaining columns

- Primary keys are unique, but could be tuples.
- Examples: SSN, ProductIDs, …

Primary Key

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Foreign keys: the column or sets of columns that reference primary keys in other tables.

More later when we see SQL.
Stay tuned!

Primary Key

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Foreign Key

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More File Formats

Structure

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- **More File Formats**

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Demo Slides

TSV: Tab Separated Values

Another common table file format.

- Records are delimited by a newline: `'\n'`, `"\r\n"`
- **Fields** are delimited by `'\t'` (tab)

`pd.read_csv`: Need to specify `delimiter='\t'`

Issues with CSVs and TSVs:

- Commas, tabs in records (use `quotechar` parameter)
- Quoting
- ...

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Demo Slides

JSON: JavaScript Object Notation

Berkeley covid cases by day (City of Berkeley)

A less common file format.

- Very similar to Python dictionaries
- Strict formatting ”quoting” addresses some issues in CSV/TSV
- **Self-documenting**: Can save metadata (data about the data) along with records in the same file

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Demo Slides

JSON: JavaScript Object Notation

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- **Self-documenting**: Can save metadata (data about the data) along with records in the same file

Issues

- Not rectangular
- Each record can have different fields
- Nesting means records can contain tables – complicated

Reading a JSON into pandas often requires some EDA.

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Are the data in a standard format or encoding?

- Tabular data: CSV, TSV, Excel, SQL
- Nested data: JSON or XML

Are the data organized in **records** or nested?

- Can we define records by parsing the data?
- Can we reasonably un-nest the data?

Does the data reference other data?

- Can we join/merge the data?
- Do we need to?

What are the **fields** in each record?

- How are they encoded? (e.g., strings, numbers, binary, dates …)
- What is the type of the data?

Structure -- the "shape" of a data file

Granularity -- how fine/coarse is each datum

Scope -- how (in)complete is the data

Summary

You will do the most data wrangling when analyzing the structure of your data.

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Scope and Temporality

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Scope

Will my data be enough to answer my question?

- **Example**: I am interested in studying crime in California but I only have Berkeley crime data.
- **Solution**: collect more data/change research question

Is my data too expansive?

- **Example**: I am interested in student grades for Data200 but have student grades for all Data Science classes.
- **Solution: Filtering** ⇒ Implications on sample?
  - If the data is a sample I may have poor coverage after filtering

Does my data cover the right time frame?

- Which brings us to **Temporality**

"Scope" questions are defined by your question/problem and inform if you need better-scoped data.

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Temporality

Data changes – when was the data collected/last updated?

Periodicity — Is there periodicity? Diurnal (24-hr) patterns?

What is the meaning of the time and date fields? A few options:

- When the "event" happened?
- When the data was collected or was entered into the system?
- Date the data was copied into a database? (look for many matching timestamps)

Time depends on where! (**time zones** & daylight savings)

- Learn to use `datetime` Python library and Pandas `dt` accessors
- Regions have different datestring representations: 07/08/09?

Are there strange null values?

- E.g., **January 1st 1970**, January 1st 1900…?

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Temporality: Unix / POSIX Time

Time measured in seconds since **January 1st 1970 UTC**

- Minus leap seconds …

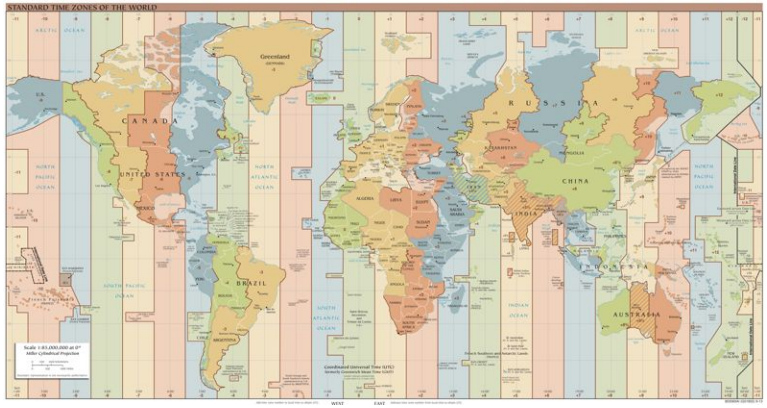
UTC is Coordinated Universal Time

- International time standard
- Measured at 0 degrees latitude
  - Similar to Greenwich Mean Time (GMT)
- No daylight savings

Time Zones:

- San Francisco (**UTC-7**) with daylight savings

Jun 27, 2023 5:00pm PDT
1687910400



https://en.wikipedia.org/wiki/Coordinated_Universal_Time

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Faithfulness (and Missing Values)

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Faithfulness: Do I trust this data?

Does my data contain **unrealistic or “incorrect” values**?

- Dates in the future for events in the past
- Locations that don’t exist
- Negative counts
- Misspellings of names
- Large outliers

Does my data violate **obvious dependencies**?

- E.g., age and birthday don’t match

Was the data **entered by hand**?

- Spelling errors, fields shifted …
- Did the form require all fields or provide default values?

Are there obvious signs of **data falsification**?

- Repeated names, fake looking email addresses, repeated use of uncommon names or fields.

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Signs that your data may not be faithful (and proposed solutions)

Truncated data

Early Microsoft Excel limits: 65536 Rows, 255 Columns

Duplicated Records or Fields

Identify and eliminate (use primary key).

Spelling Errors

Apply corrections or drop records not in a dictionary

Units not specified or consistent

Infer units, check values are in reasonable ranges for data

Time Zone Inconsistencies

Convert to a common timezone (e.g., UTC)

- Be aware of consequences in analysis when using data with inconsistencies.
- Understand the potential implications for how data were collected.

Missing Data???

Examples

```
"  "          1970, 1900
0, -1         NaN
999, 12345    Null
```

NaN: “Not a Number”

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Missing Data/Default Values: Solutions

A. Drop records with missing values

- Probably most common
- **Caution**: check for biases induced by dropped values
  - Missing or corrupt records might be related to something of interest

B. Keep as NaN

C. Imputation/Interpolation: Inferring missing values

- **Average Imputation**: replace with an average value
  - Which average? Often use closest related subgroup mean.
- **Hot deck imputation**: replace with a random value
- **Regression imputation**: replace with a predicted value, using some model
- **Multiple imputation**: replace with multiple random values.

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(beyond this course)

Choice affects bias and uncertainty quantification (large statistics literature)

Essential question: why are the records missing?

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Demo: Mauna Loa CO2

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Demo Slides

What are our Variable Feature Types?

EDA step:
Understand what each record, each feature represents

First, **read file description**:

- All measurement variables (**`average`**, **`interpolated`**, **`trend`**) are monthly mean CO2 monthly mean mole fraction
  - i.e. monthly average CO2 ppm (parts per million)
  - Computed from daily means
- **`#days`**: Number of daily means in a month (i.e., # days equipment worked)

What variables define the first three columns?

- Year, month, and date in decimal

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Demo Slides

The Search for the Missing Values

EDA step:
Hypothesize why these values were missing, then use that knowledge to decide whether to drop or impute missing values

From file description:

- **`-99.99`**: missing monthly average **`Avg`**
- **`-1`**: missing value for **`# days`** that the equipment was in operation that month.

Which approach?

- Drop missing values
- Keep missing values as NaN
- Impute

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Summary: Dealing with Missing Values

Mauna Loa Observatory CO2 levels (NOAA)

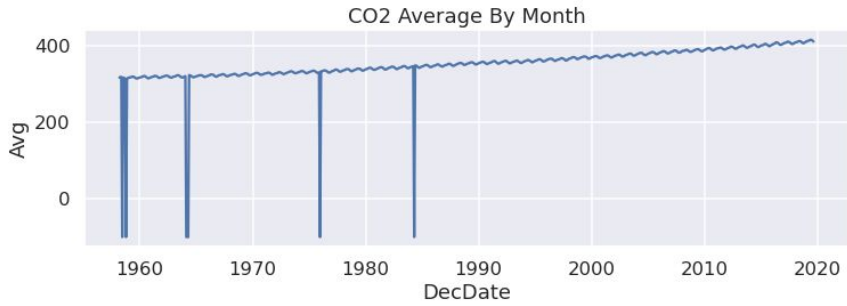
`-99.99`: missing monthly average **`Avg`**

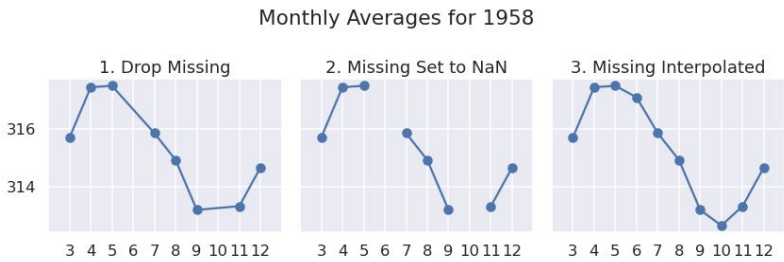
Option A: Drop records

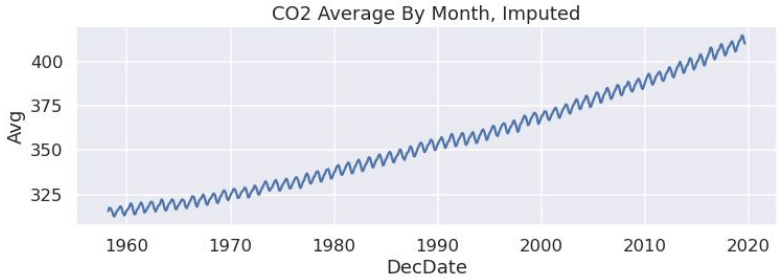
Option B: NaN missing values

Option C: **Impute** using interpolated column **`Int`**

All 3 are probably fine since few missing values, but we chose Option 3 based on our EDA.







With **numeric data**, you generally wrangle as you do EDA.

With **text data**, **wrangling is upfront** and requires new tools: **Python string manipulation** and **regular expressions**.

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Summary: How do you do EDA/Data Wrangling?

Examine **data and metadata**:

- What is the date, size, organization, and structure of the data?

Examine each **field/attribute/dimension** individually

Examine **pairs of related dimensions**

- Stratifying earlier analysis: break down grades by major …

Along the way:

- **Visualize**/summarize the data
- **Validate assumptions** about data and collection process. Pay particular attention to when data were collected.
- Identify and **address anomalies**
- Apply data transformations and corrections (next lecture)
- **Record everything you do**! (why?)
    - Developing in Jupyter Notebooks promotes reproducibility of your own work.

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Aside: An update to the Mauna Loa Dataset

https://gml.noaa.gov/ccgg/trends/data.html

```
# NOTE: Due to the eruption of the Mauna Loa Volcano, measurements from Mauna Loa Observatory
# were suspended as of Nov. 29, 2022. Observations starting in December 2022 are from a site
# at the Maunakea Observatories, approximately 21 miles north of the Mauna Loa Observatory.
#
```

NPS

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